Netflix Prize Dataset

Recommendation Systems

for Personalized Content Discovery

Collaborative Filtering · Matrix Factorization · Ranking Evaluation

574K	66.9K	277	98.7%
Ratings	Users	Movies	Sparse

Problem Overview

The Challenge

- 100M+ ratings, 480K users, 17,770 movies
- Matrix is 98.7% EMPTY — most movies never rated by each user
- Goal: predict what a user would rate an unseen movie
- And generate a personalised Top-10 list

Our Approach

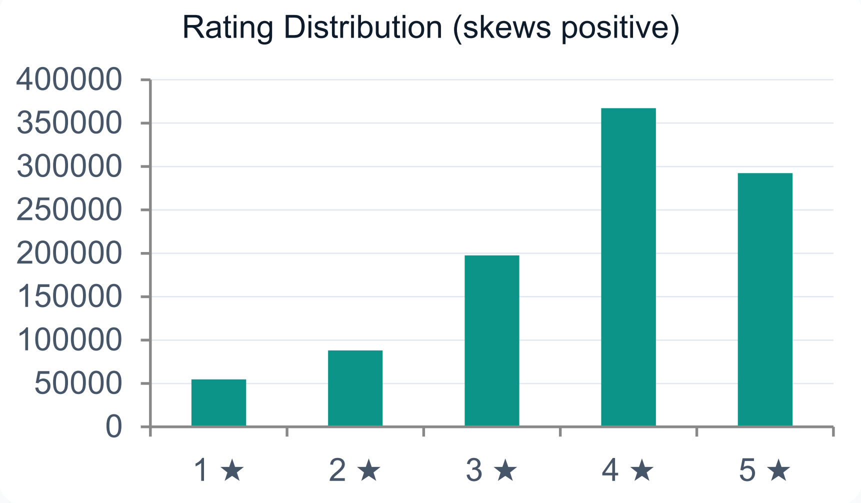
SVD Matrix Factorization — learns latent user & item factors

Item-KNN Item-Based CF — similarity-weighted predictions

Metrics RMSE + MAP@10 (mandatory) + Precision/Recall/NDCG

Explain "Because you liked X" — KNN similarity explanations

Exploratory Data Analysis — Key Findings



Positive Skew

Mean = 3.64, Median = 4.0
Users rate movies they chose to watch

Long-Tail Users

Top 1% users = 7.5% of ratings
Median user has only 2 ratings

Matrix Sparsity:

98.72 %

of the user–item matrix is empty. Only 1.28% of all possible (user, movie) pairs have a rating.
This is why collaborative filtering is hard — models infer from a tiny slice of observed interactions.

Methodology

Two complementary approaches — accuracy vs. interpretability

MODEL A

SVD — Matrix Factorization

- Decomposes rating matrix $R \approx U \times \Sigma \times V^T$
- Learns dense latent vectors (100 factors) for users & movies
- SGD optimisation, 20 epochs, lr=0.005, reg=0.02
- Handles sparsity naturally via low-rank approximation
- Seed=42 for reproducibility

MODEL B

Item-Based KNN (CF)

- Cosine similarity over co-rated user pairs
- K=40 nearest neighbour movies per prediction
- min_support=3 (needs ≥3 co-raters for valid sim)
- KNNWithMeans: subtracts per-item mean to reduce bias
- Highly interpretable — powers explainability layer

Pipeline: Load → Filter (≥5 ratings per user/movie) → 80/20 split → Train → Evaluate → Retrain on 100% → Recommend

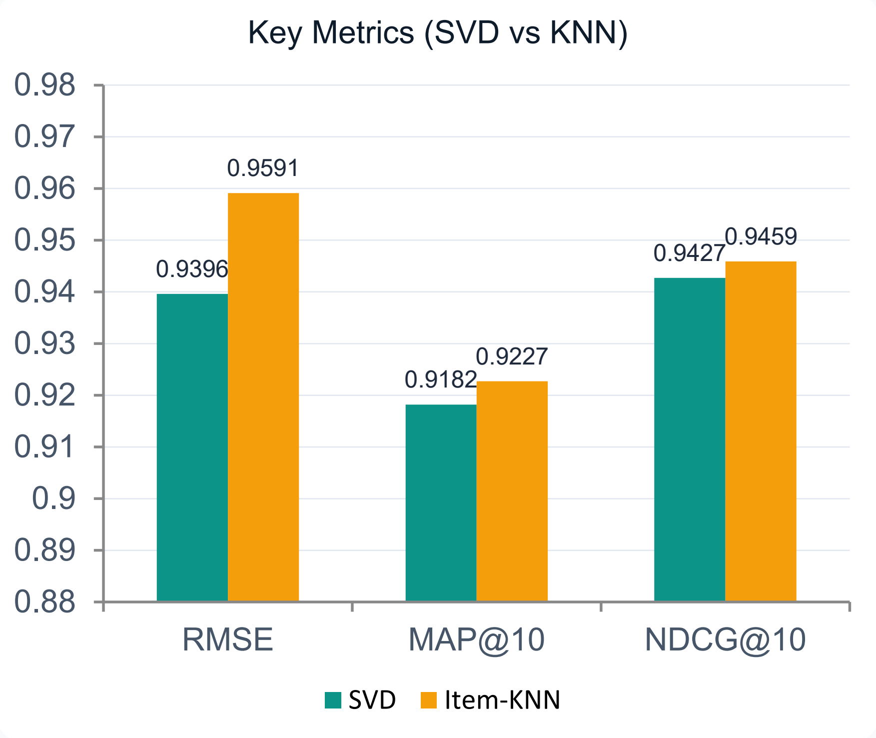
Experimental Results

SVD (Best Model)

0.9396	0.9182	0.7405
RMSE ↓	MAP@10 ↑	MAE ↓

Item-KNN

0.9591	0.9227	0.7473
RMSE ↓	MAP@10 ↑	MAE ↓



Model Comparison

Criterion	SVD (Matrix Factorization)	Item-Based KNN
RMSE	**0.9396 ✓ Better**	0.9591
MAP@10	0.9182	**0.9227 ✓ Better**
Training Time	Fast (SGD, seconds)	Slow (sim matrix)
Memory	Low (factor vectors)	High (N×N matrix)
Interpretability	Low (latent space)	**High ✓ — explains recs**
Cold-Start	Partial (global bias)	Weak without history
Scalability	Excellent — O(factors)	Limited — O(items^2)

Verdict: SVD wins on RMSE; KNN wins on MAP@10 and interpretability. SVD chosen as primary recommendation engine.

Recommendation Examples

Success Case — User 387418 (270 ratings in filtered dataset) · Top-10 generated by SVD final model

Top-5 Recommendations

1. **Monster-in-Law: Bonus Material**
   2005 ★ 3.54
2. **Main Hoon Na**
   2004 ★ 3.54
3. **LOTR: Return of the King Extended**
   2003 ★ 3.54
4. **Pitch Black**
   2000 ★ 3.53
5. **Brokeback Mountain**
   2005 ★ 3.53

Why these?

Monster-in-Law recommended because you liked 'I Spy' and 'Duplex (Widescreen)'

Main Hoon Na recommended because you liked 'Duplex (Widescreen)' and 'I Spy'

LOTR Extended recommended because you liked 'I Spy' and 'Duplex (Widescreen)'

Key Insights & Future Work

What We Found

- **Positive rating bias** Selection bias — users watch movies they expect to like; mean = 3.64
- **Long-tail dominance** Top 1% power users drive 7.5% of ratings; models must not overfit to them
- **SVD wins on accuracy** RMSE 0.9396 vs KNN 0.9591 — latent factor approach handles sparsity better
- **KNN wins on insight** "Because you liked X" explanations work perfectly with item similarity scores
- **Coverage = 100%** Both models can generate recommendations for every user in the test set

Future Improvements

- **SVD++** +5–10% RMSE — implicit feedback (views, browsing)
- **Neural CF** Deep network replaces dot-product — non-linear patterns
- **Temporal Model** Model taste drift over time (TimeSVD++)
- **Hybrid** SVD + KNN + content features (genre, actors)
- **Full 100M Dataset** Richer representations from all 100M+ ratings

Final: SVD RMSE=0.9396 | SVD MAP@10=0.9182 | KNN MAP@10=0.9227 | Coverage=100% | Best Model: SVD (Matrix Factorization)